Project Proposal: Algorithmic Sequence Ordering for Multi-View 3D Super-Resolution

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Topic, Motivation, and Course Relevance

We propose to study **algorithmic sequence ordering for unordered multi-view images**. The application context is 3D super-resolution, but the core of the project is an algorithm-design problem: given a set of unordered views, construct one or more smooth image sequences so that adjacent frames are as similar as possible. This sequence construction problem is directly related to course topics such as graph modeling, greedy algorithms, dynamic programming, shortest-path-style optimization, approximation behavior, complexity analysis, and empirical evaluation of algorithmic trade-offs.

The motivating paper, *Sequence Matters: Harnessing Video Models in 3D Super-Resolution*, shows that video super-resolution models can be reused for 3D super-resolution if multi-view images are first arranged into video-like sequences. However, the ordering step is not simply a data pre-processing detail. It is the key computational decision: poor ordering creates abrupt frame transitions and damages the downstream output, while better ordering improves temporal smoothness and reconstruction quality. Therefore, our project will de-emphasize low-level image pre-processing and 3D reconstruction engineering, and instead focus on the design, analysis, and comparison of sequence-ordering algorithms.

Computational Problem

The input is an unordered set of low-resolution posed images $I = \{(I_i, P_i)\}_{i=1}^{N}$, where I_i is an image and P_i is its camera pose. We represent each image as a vertex in a complete weighted graph $G = (V, E)$, $|V| = N$, where each edge weight $C[i, j]$ measures the dissimilarity between two views.

The output is an ordered sequence $S = (I_{\pi_1}, I_{\pi_2}, \ldots, I_{\pi_N})$, or a set of adaptive subsequences $\{S^{(j)}\}_{j=1}^{M}$. The surrogate objective is $\min_\pi \sum_{t=1}^{N-1} C[\pi_t, \pi_{t+1}]$, which can be viewed as a shortest Hamiltonian-path-style problem on a weighted graph. This formulation connects the project directly to graph modeling, greedy algorithms, dynamic programming, and complexity analysis.

Baseline Algorithm: Greedy Nearest-Neighbor Ordering

The baseline algorithm from the paper is a greedy nearest-neighbor method. Starting from an initial image I_s, it repeatedly appends the closest unvisited image, i.e., $\pi_{t+1} = \arg\min_{k \notin \{\pi_1, \ldots, \pi_t\}} C[\pi_t, k]$.

With a precomputed similarity matrix, the algorithm requires $O(N^2)$ time and $O(N^2)$ memory. It is simple and efficient, but it is not globally optimal. The greedy strategy may consume locally good neighbors too early and leave distant views for the end, causing large late-stage jumps. This makes it a useful case study for analyzing the strengths and limitations of greedy algorithms.

Adaptive Subsequence Construction

Instead of forcing all images into one long sequence, we will also reproduce adaptive-length subsequence construction. Starting from each image, the algorithm greedily extends a subsequence but stops when the best next edge exceeds a threshold, i.e., $\min_{k \notin S} C[S_t, k] > \epsilon$.

This converts the problem from one global path into multiple locally smooth paths. A multi-threshold version first uses a strict threshold to obtain highly smooth subsequences, then gradually relaxes the threshold to improve coverage. This part studies the algorithmic trade-off between sequence smoothness and image coverage.

Proposed Algorithmic Improvements

To strengthen the connection with the course, we will compare the greedy baseline with more principled alternatives.

1. Dynamic programming for small scenes. For small N, we will implement a Held–Karp-style dynamic programming method for the shortest Hamiltonian path: $DP[S][j] = \min_{i \in S, i \neq j} DP[S \setminus \{j\}][i] + C[i, j]$. This provides an exact or near-exact reference for small subsets, allowing us to quantify how far the greedy method is from the optimal ordering. The expected complexity is $O(N^2 2^N)$, so it will only be used for small scenes or sampled subsets.

2. Windowed dynamic programming and local search. For larger image sets, we will first obtain a rough sequence using greedy ordering and then re-optimize local windows by DP. We will also test a simple 2-opt-style local search, where a sequence segment is reversed if replacing two edges reduces the total path cost. These methods provide practical compromises between global optimality and computational feasibility.

3. Low-resolution sub-pixel consistency loss. Besides sequence ordering, we will include a lightweight reconstruction-side improvement inspired by the paper's low-resolution constraint. Given the rendered high-resolution image $R_\theta(P_i)$ and the original low-resolution input I_i, we add $\mathcal{L}_{\mathrm{LR}} = \| \downarrow R_\theta(P_i) - I_i \|_1 + \alpha \mathcal{L}_{\mathrm{DSSIM}}(\downarrow R_\theta(P_i), I_i)$, where $\downarrow$ denotes bicubic downsampling. The full loss is $\mathcal{L} = \lambda \mathcal{L}_{\mathrm{HR}} + (1-\lambda)\mathcal{L}_{\mathrm{LR}}$. This term discourages VSR-generated high-frequency artifacts that are inconsistent with the observed LR images, while remaining secondary to the main algorithmic study.

Distance Functions and Data Structures

The main data structure is the similarity matrix $C \in R^{N \times N}$. We will implement and compare the following edge weights:

- **Pose distance**: Euclidean distance between camera centers plus angular distance between viewing directions.
- **Feature distance**: ORB descriptor matching distance using cross-checked Hamming distance.
- **Hybrid distance**: a weighted combination of pose and feature distances.

Feature extraction is treated only as a way to define graph edge weights. The project will not focus on image-processing tricks or heavy pre-processing. The algorithmic focus remains on how different edge definitions affect the behavior of greedy, DP, and local-search ordering methods.

Project Scope and Evaluation

To keep the project feasible and course-focused, we will use a small subset of NeRF Synthetic scenes, such as Lego and Chair, with bicubic downsampling to generate low-resolution inputs. We will compare naive ordering, greedy nearest-neighbor ordering, adaptive subsequences, greedy plus local search, and DP/windowed-DP ordering on small subsets.

The evaluation will mainly focus on algorithm-level metrics, including total path cost, maximum adjacent-frame jump, adjacent-distance distribution, subsequence coverage, runtime, and memory usage. If time permits, we will use a pretrained VSR model and 3DGS as a downstream validation stage, measured by PSNR, SSIM, and LPIPS. The final report will emphasize graph formulation, greedy behavior, DP/local-search comparison, complexity analysis, and failure-case analysis.

Team Information and Contribution Plan

Xu Qihan: implement pose parsing, pose-based edge weights, and algorithm-level smoothness evaluation. **Zhong Yiming**: implement ORB feature extraction, descriptor matching, and feature-based edge weights. **Wang Luo**: reproduce greedy ordering and adaptive subsequence generation, including multi-threshold selection. **Feng Xiang**: implement DP/windowed-DP/local-search variants, organize the proposal and final report.**Cheng Wenyu**: prepare experiments, presentation slides, and result visualization.

References

[1] C. Wang et al. NeRF-SR: High Quality Neural Radiance Fields Using Supersampling. ACM MM, 2022.

[2] Y. Shen et al. SuperGaussian: Repurposing Video Models for 3D Super Resolution. arXiv, 2024.

[3] H. Ko et al. Sequence Matters: Harnessing Video Models in 3D Super-Resolution. AAAI, 2025.